An Optimized Edge-Focused Siamese Network for Monitoring New Illegal Buildings Using Satellite Images

Haode Du, Zhou Huang, and Yi Zhang

***Abstract*— Illegal construction is a common problem often encountered by cities with rapid development, which is hard to deal with for multiple reasons. Although these illegal buildings are primarily defined by laws and regulations, they still have physical characteristics in common that makes them identifiable. In this study, we propose an illegal building monitoring method based on satellite images and deep learning techniques, named illegal building monitoring network (IBMNet), to improve the data collection capacity for monitoring new illegal buildings. IBMNet is an end-to-end pixelwise segmentation network with two flows: the Segment Flow, which includes a Siamese encoder and an attention fusion module (AFM), and the Edge Flow, which uses Gated Convolutional Layers to extract edge information. We implement and evaluate our model in China, a country with fast development and struggling with illegal buildings. In addition to the conventional metrics, we propose a set of specialized metrics to evaluate the model's ability to discriminate illegal buildings and legal buildings (OA_B, $F1_B$, and IoU_B). The model achieves great results on the illegal building monitoring dataset (IBMD) with an $F1$ score of 0.7990 and IoU_B of 0.7449, showing its great ability in detecting illegal buildings in various scenarios and distinguishing them from legal buildings. Compared to the existing methods based on urban database, IBMNet has a higher time resolution and a larger space coverage, making it more accessible in data and cost-effective for governments. The proposed method is also promising in other cities and countries with similar problems.**

***Index Terms*— Change detection, edge focus, illegal building monitoring, Siamese network.**

I. Introduction

DURING periods of rapid development, it is common for cities to face challenges related to irregularities and disordered development [1], [2]. Illegal construction represents one such challenge. Illegal buildings are commonly defined as buildings that violate local construction regulations or urban planning laws [3], [4]. Accepting this definition, illegal buildings may include structures that lack the necessary permits, have been erected on protected land, exceed height or size limits, or have been constructed in a manner that does not comply with safety standards. The global impact of illegal buildings is significant [3], [5], particularly as many individuals migrate from rural areas to urban centers, exacerbating the issue of informal construction in cities with fast-paced development [6], [7].

However, the surveillance of illegal construction presents significant challenges to authorities for several reasons. First, illegal buildings have unique characteristics that make effective supervision difficult. Illegal buildings can be sporadic and widely dispersed, spanning across urban, suburban, and rural areas, thereby requires widespread surveillance [8]. Moreover, illegal construction is known for its rapid pace, with buildings erected in a single night and 2–3 floors constructed per day [9], which increases the difficulty of timely supervision. Second, the legislation related to illegal construction is not unified and perfect [4], [6]. The existence of legislative loopholes provides many gray areas for the construction of illegal buildings, thereby increasing the difficulty of enforcement against them. Third, compounding this difficulty is the government's inability to acquire accurate data on a large scale, which is due to the lack of technical–administrative structures and reliable methods for data capture [3]. Inconsistencies between the central government's policies and local government practices [1], [2], [7], [10], as well as conflicts among political parties or groups [7], [11], further impede efforts to acquire information and act against illegal construction. Finally, the cost of demolishing illegal buildings that have been standing for an extended period [3], along with the compensation and staffing required to persuade residents, places a significant burden on local governments. In short, there are still many problems to be solved in the supervision of illegal construction.

The development of remote sensing and computer vision technologies offers a promising solution to partially solve the problem of monitoring illegal buildings from the aspect of data collection. Recently, the integration of very-high-resolution (VHR) satellite images and deep learning techniques has been widely implemented to solve urban problems [12], [13], [14]. Similarly, this method can be applied to monitor illegal buildings. VHR images with high spatial and temporal resolutions facilitate the timely and accurate detection of illegal buildings. With deep learning neural networks, visual information can be extracted and comprehended to identify illegal buildings from complex backgrounds on large scales.

Although illegal buildings are primarily defined by urban planning laws and construction permits, they often share

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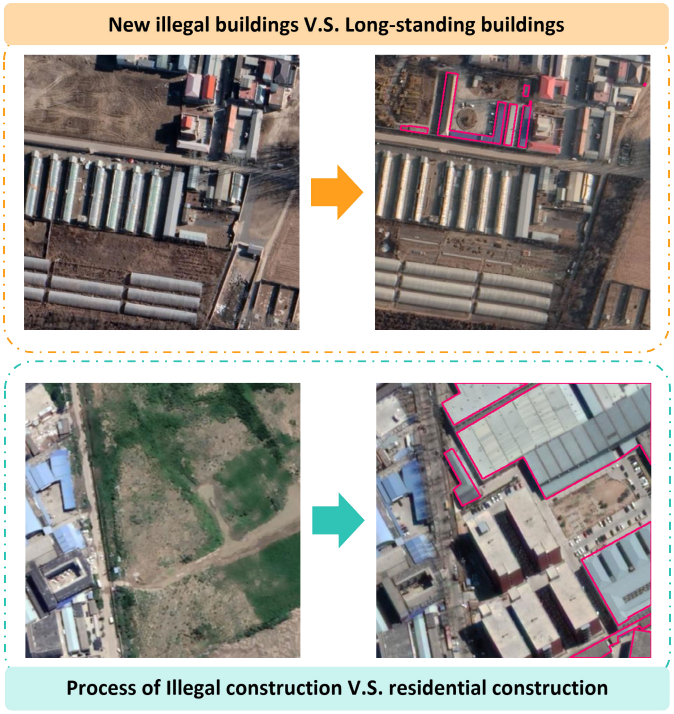


Fig. 1. Scenarios of using deep learning-based change detection method in illegal building monitoring. The illegal buildings are outlined in pink.

similar physical characteristics such as shape, texture, and color [5], [8]. Specifically, these structures are typically characterized by lower heights and have homogenous roof colors, which distinguish them from formal residential structures (see Fig. 1). It is therefore feasible to speculate that neural networks can detect deep semantic features of illegal constructions in remote sensing imagery. Meanwhile, many studies have recognized the potential of change detection for monitoring illegal constructions [15], [16], [17], while few have explored its detailed practical applications. Change detection enables the timely identification of newly built illegal buildings with high temporal resolution satellite imagery. This approach is particularly beneficial in that it can easily distinguish newly constructed illegal buildings from similar long-standing constructions in factories or farms, such as warehouses and greenhouses (see Fig. 1), thereby narrowing the scope of inspection.

Previous studies have explored methods for detecting and monitoring illegal constructions using VHR satellite images while still needs the support of urban databases [18], [19]. However, both methods are limited by the quality and update frequency of urban databases, which rely on communication between government sectors [20], [21] and may be affected by the inconsistency mentioned above across different levels of government. Given that illegal buildings are constructed rapidly [9], the existing change detection methods based on urban databases are insufficient in temporal resolution. The urban databases also limit the spatial coverage of these methods, as they cannot effectively cover suburban areas with imperfect statistics.

In this work, we aim to offer a promising solution for large-scale data collection and timely monitoring of illegal buildings using change detection of remote sensing images combined with state-of-the-art deep learning methods. To overcome the limitations of urban databases, we use the satellite images as the sole data source, decoupling illegal construction monitoring from urban databases and then enabling more flexible and frequent monitoring. We demonstrate the effectiveness of our method through implementation and evaluation in China, a country that has experienced rapid urbanization and economic growth in the past four decades, yet still suffers from the problem of illegal construction.

We present a novel deep learning network, named illegal building monitoring network (IBMNet), for detecting newly constructed illegal buildings through change detection in bitemporal VHR satellite imagery. IBMNet has a two-flow structure comprising the Segment Flow and Edge Flow, which enables the extraction of pixelwise illegal building contours from complex contexts. The Segment Flow includes a dual-branch Siamese encoder that extracts multiscale representations from the images. The attention fusion module (AFM) fuses the multiscale features extracted by the encoder, combining the information from two images. The Edge Flow utilizes gated convolution layers (GCLs) to extract representations of the boundaries of illegal buildings. The Fusion Block integrates the semantic features from the Segment Flow and the edge features from the Edge Flow to generate the final pixelwise illegal building identification map. To evaluate IBMNet's performance, we constructed the illegal building monitoring dataset (IBMD) using data provided by the Beijing municipal government, including 5230 pairs of satellite images. Moreover, special metrics are proposed to evaluate the model's ability in discriminating construction of illegal buildings and legal buildings without the information about legal buildings. Comparative and ablation experiments were conducted on IBMD to demonstrate the effectiveness and accuracy of our network in illegal building monitoring. Our contributions are summarized as follows.

1) We propose a novel two-flow change detection network, IBMNet, for illegal building monitoring based on a Siamese network and Gated Convolutional Layers. Our Edge Flow is designed to refine the segmentation boundary and extract small illegal buildings from complex backgrounds. We also introduce the AFM to integrate features extracted from bitemporal images. Our work is the first to apply the state-of-the-art edge-focusing method to the context of illegal building monitoring, increasing accuracy while reducing the data requirement.
2) Our method enhances the capture of information regarding illegal buildings monitoring by relying solely on satellite imagery, effectively avoiding the limitations of urban databases. Unlike urban databases, which are constructed hierarchically through government statistics obtained at different administrative levels, our method generates unified results from satellite images. Thanks to the high temporal resolution of remote sensing imagery, the detection frequency is markedly improved. This enables the government to regulate illegal construction proactively and suppress it at an early stage, thereby reducing the cost of government surveillance.

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Additionally, our proposed method has the advantage of encompassing suburban areas that are often overlooked in databases.

3) To evaluate the effectiveness of our method in distinguishing illegal buildings from legal buildings without explicitly learning the features of legal buildings, we introduce a set of specialized metrics. By measuring the binary segmentation capability of the model for illegal and legal constructions, these metrics assess the discriminative ability of the model.
4) We have preliminarily validated the feasibility of monitoring illegal buildings solely relying on the remote sensing imagery, which presents a progressive step toward efficient and expansive supervision and data collection of illegal construction.

The rest of the article is organized as follows. Section II introduces the related works in illegal building monitoring and remote sensing change detection methods based on deep learning. Section III demonstrates network architecture, design of loss functions, and dataset of our methods. Section IV reports the experiment details and the results of comparative experiments. Section V introduces the result of ablation studies and discusses the effectiveness of our network. Section VI summarizes the whole work and provides an outlook on the future.

II. Related Works

A. Illegal Building Monitoring

Recently, the rapid evolvement of computer vision technologies combined with artificial intelligence and deep neural network architecture provides automated solutions for a wide range of conventional image tasks, including industrial defect detection, postphotographic adjustments, and remote sensing interpretation. In the field of illegal building monitoring, the real-time images recorded by widely distributed surveillance cameras have been used by researchers to monitor newly built illegal buildings [22], [23]. Chen et al. [22] compared the corrected surveillance images from different periods and detected the newly built illegal buildings from a comprehensive view of color space and texture. Wang et al. [23] explored a method to identify unauthorized structures based on surveillance images in different climates and urban scenarios.

The development of remote sensing technology has enabled the acquisition of images of large areas with higher spatial and temporal resolutions. The enhancement of image qualities makes it possible to conduct large-scale automatic identification of illegal buildings from satellite images. Most existing research extracted features and identified illegal buildings based on the combination of the city database and remote sensing images. The methods of these studies can be typically attributed to two categories.

One is to extract the building boundaries and subsequentially compare the extracted features with the city database to the validity of the building. Ostankovich and Afanasyev [18] used GoogLeNet to detect building boundaries and distinguish illegal buildings using cadastral maps. Liu et al. [24] identified illegal buildings in river areas via DeepLabv3+ based on the knowledge that all buildings that emerge near the river are illegal due to no-building areas aside from the riverway in urban planning.

The other one is the change detection method. The change maps of buildings are created by conventional or deep learning change detection methods. The change maps are then compared with the urban construction plans to determine the locations of illegal buildings. Khalilimoghadama et al. [19] proposed a fuzzy XOR operation method to detect building changes and then determine the legality of the construction and modification of the building by the construction permits. Chaabane et al. [25] detected illegal buildings by constructing satellite image time series and proposed a framework to measure spatiotemporal similarity for building change detection. Li and Tan [21] constructed a building inventory database using ground-based remote sensing and detected illegal buildings by combining multiple urban databases.

B. Change Detection Based on Deep Learning

In recent years, deep learning methods have been more favored in remote sensing analysis compared to conventional image analysis methods [26] for their advantages in learning deep semantic features from remote sensing imagery [27]. Various deep learning techniques have been applied in the field of remote sensing change detection, such as U-Net [28], deep belief network (DBN) [29], generative adversarial network (GAN) [30], and recurrent neural network (RNN) [16], [31], achieving excellent results. The deep learning methods for change detection could be divided into two major types: supervised learning and unsupervised learning.

1) Supervised Learning: Supervised change detection methods can be categorized into two frameworks: single stream and double stream. Single-stream frameworks utilize data processing techniques to merge two or more sets of data, which are then subjected to feature learning via deep learning models. Data fusion is commonly achieved through change analysis methods (e.g., change vector analysis (CVA) [32], differencing by log-ratio [33], [34], [35], and change measures [36], [37]) or channel concatenations [29], [38]. In contrast, the double-stream framework typically employs a Siamese structure comprising two branches with the same structure that generate two comparable features for change detection. The double-stream framework can further be divided into pure Siamese structures [28], [39] with shared weights and pseudo-Siamese structures [40] with different weights. Popular mainstream networks used in both frameworks are convolutional neural networks (CNNs), such as Visual Geometry Group (VGG) [39], [41], residual neural network (ResNet) [42], [43], U-Net [17], [24], [28], and InceptionNet [44]. RNNs, including long short-term memory (LSTM) [45], are also widely used for their ability to deal with temporal sequence data. Besides, generative methods like GANs [46], [47] are emerging as promising approaches for change detection. Recently, as the transformer architecture has shown great potential in natural language processing, it has also been applied to remote sensing image change detection [48]. Various approaches have been developed in this context, including

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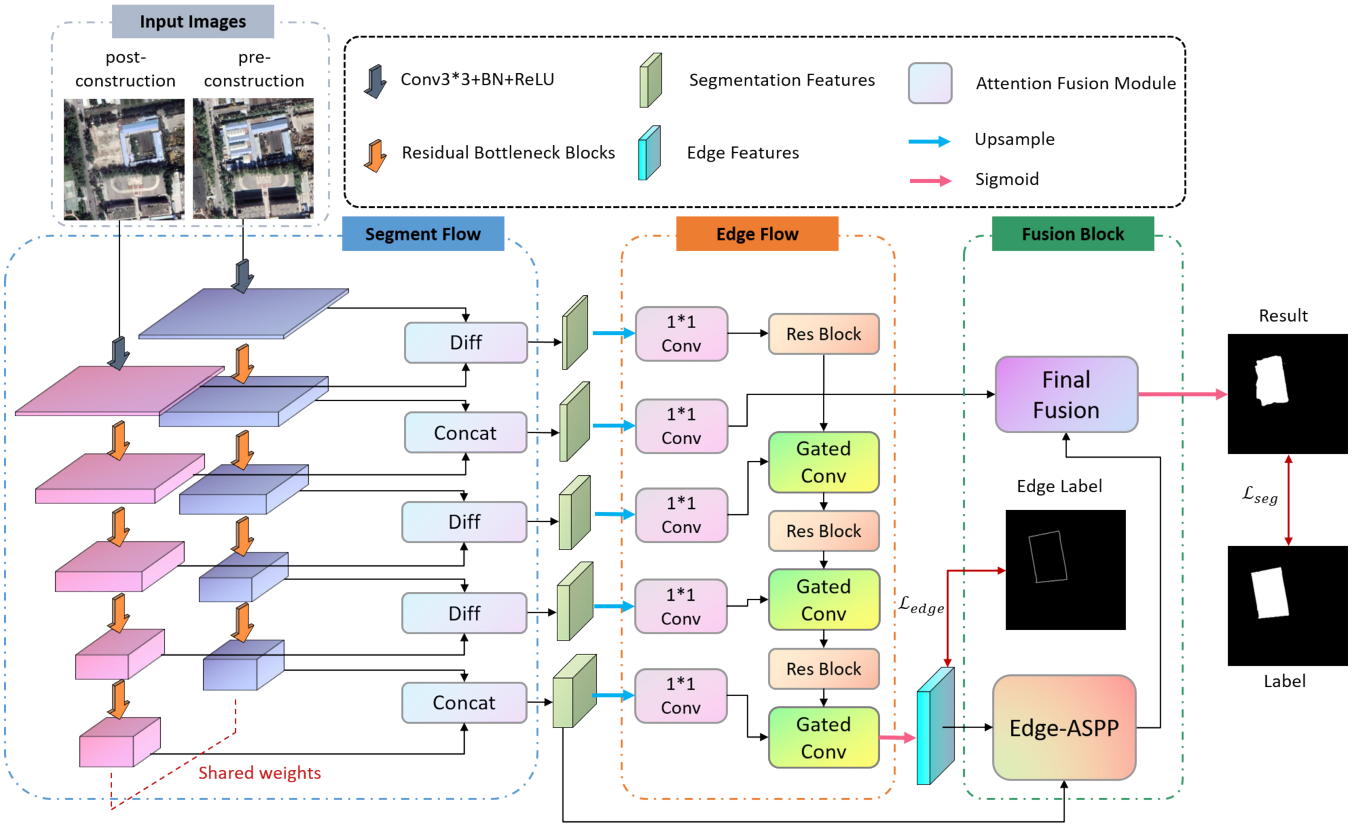


Fig. 2. Network architecture of IBMNet. The IBMNet includes Segment Flow using Siamese encoder and AFM, Edge Flow using GCLs, and Fusion Block to fuse features from different flows. The types of fusion operators used by AFM are annotated inside their blocks. Pixelwise edge labels and segmentation are used to supervise the network. L_{seg} and L_{edge} represent the loss function of testing the change map and the loss function of edge extraction.

pure transformer-based methods [49], [50] and hybrid methods that integrate transformer and CNN [48], [51].

2) Unsupervised Learning: Unsupervised learning has also been widely used in building change detection because large amounts of labeled image pairs for supervised training are difficult to generate and have substantial finance and time costs. Such methods were usually implemented with DBN and AutoEncoder (AE). AEs can be employed for feature dimensionality reduction and outperform principal component analysis (PCA) due to the robust feature learning abilities of neural networks. As such, they are extensively used as feature extractors in change detection. The most commonly used AEs include stacked AEs [52], stacked denoising AEs [53], stacked fisher AEs [54], sparse AEs [55], and fuzzy AEs [56]. Gong et al. [34] proposed a change detection method that integrates Sparse AE (SAE), CNN, and unsupervised clustering. DBN, a generative graphical model which learns probabilities to reconstruct its inputs, also works as feature extractor in many unsupervised change detection works [57], [58]. Also, some works have used the deep Boltzmann machine (DBM), a graph model similar to DBN but undirected, as the feature extractor [59]. Apart from DBNs and AEs, Tang et al. [60] introduced graph convolutional networks and metric learning to capture contextual patterns and extract changed and unchanged areas of buildings without labels. Additionally, transformer-based networks are also used in unsupervised change detection [61], [62], [63]. In these studies, transformers are employed to capture the spatial and temporal information of the remote sensing images [62], as well as to detect difference between images [61], [63].

III. Methodology

A. Network Architecture

As shown in Fig. 2, IBMNet is a Siamese neural network that takes two RGB images as inputs and generates outputs of a binary change map. IBMNet is constituted by three parts: Segment Flow, Edge Flow, and Fusion Block. The Segment Flow includes a weighted-sharing encoder, adapted from ResNet101 [64] and a set of AFM modules. The Siamese encoder extracts multiple hierarchical features from bitemporal images and then sends the features to the AFM for further feature comprehension. AFM is designed based on two attention mechanisms (spatial and channel attentions), which can fuse features from two encoder branches in different approaches according to the downstream tasks. It has been regarded as more effective than simple feature stitching or differentiating. The Edge Flow uses GCL to process the edge information from hierarchical semantic features to generate clear boundaries. The Edge Flow is enforced to focus only on edge-related information by local supervision. Then, the Fusion Block integrated the deep semantic features from the Segment Flow and the edge features from the Edge Flow to generate the final binary map. In the Fusion Block, the atrous spatial pyramid pooling for edge features (Edge-ASPP)

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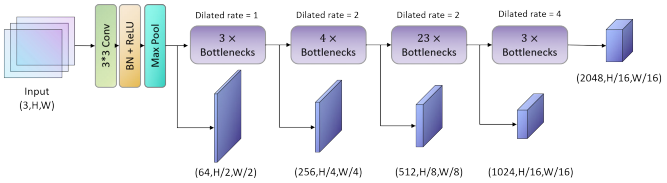


Fig. 3. Backbone of the Siamese encoder. H and W represent the height and width of the feature map. Different dilated rates are used in the encoder. As the feature maps get smaller, the dilated rate used by convolutions increases.

module integrates deep semantic features and edge features to generate multiscale features. After that, the final fusion module fuses multiscale features into the segmentation results of illegal buildings with refined boundaries.

1) Segmentation Flow:

a) Siamese encoder: The Siamese network has been widely used in several areas, including semantic segmentation [65], natural language processing [66], and unsupervised and semisupervised learning [65], [67]. Branches of the Siamese network share the same structures and weights, which enables it to compute comparable features for further change map generation that is suitable for change detection.

Our Siamese encoder uses two deep encoders adapted from ResNet101 to extract hierarchical semantic features from bitemporal satellite images (Fig. 3). The encoder outputs five features on different depths for the edge feature extraction of the Edge Flow and the multiscale feature fusion of the Fusion Block. Convolution of different dilated rates is performed to avoid the loss of detailed information due to downsampling. As the feature maps get smaller, the dilated rate increases to extract information from larger spatial extents.

b) Attention fusion module: Since the Siamese encoder extracts two sets of hierarchical comparable feature maps, it is crucial to find an effective approach to compare and fuse these features. Conventional methods, such as aggregating bitemporal feature maps by channel concatenation or elementwise differentiation, are poor in expressing semantic information [68]. To address this issue, this study proposes a new module which is the AFM. AFM focuses on both spatial and channel information and can be adapted for different types of typical fusion operations. The architecture of AFM is illustrated in Fig. 4, where F_1 and F_2 are the feature maps of the same hierarchy level generated by the Segment Flow. First, according to the future use of the feature maps, the two features are channelwise concatenated or elementwise subtracted and then passed into a 1×1 convolution layer to get the simple integrated feature map in the shape of $C \times H \times W$.

Next, the spatial and channel branches process the integrated feature to generate the spatial and channel attention weights. In the spatial attention branch, the feature map is handled by two pointwise convolution layers, as well as the subsequent batch normalization layer and the ReLU activation function. The spatial attention weights are finally obtained as follows:

$$W_s = \mathrm{BN}(\mathrm{Conv}_1(\mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_1(F_i))))) \tag{1}$$

where BN represents the BatchNorm layer and Conv_1 represents the 1×1 pointwise convolution layer. To reduce the

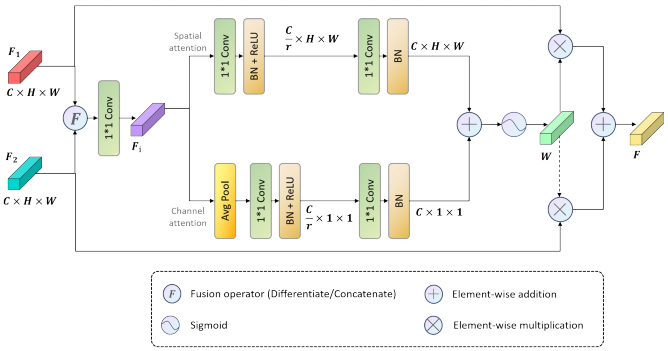


Fig. 4. AFM. C represents the channel dimension. r is the reduce factor of channels. The dashed line denotes that the weight used in multiplication is $(1-w)$.

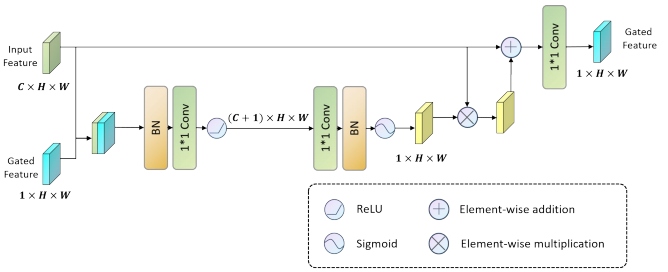


Fig. 5. Structure of GCL. C denotes the channel size of input feature from the Segment Flow.

computational cost, the number of channels is compressed in the first convolutional layer to C/r, and r is manually set to 2 in all AFMs of the IBMNet.

In the channel attention branch, we first use a global average pooling layer to get the global information of each channel. The global information is then processed by a similar procedure as for the spatial attention branch to generate the channel attention weights as below

$$W_f = \mathrm{BN}(\mathrm{Conv}_1(\mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_1(G(F_i)))))) \tag{2}$$

where G represents the global average pooling.

After the spatial attention weights and the channel attention weights are obtained, they are fused to generate the final attention weight W by the sigmoid function. The attention weight W is subsequentially used to calculate the fused feature map F with a reweighting process

$$F = (W \otimes F_1) \oplus ((1-W) \otimes F_2) \tag{3}$$

where $\otimes$ represents elementwise multiplication and $\oplus$ represents elementwise addition.

2) Edge Flow: Most illegal buildings have relatively regular shapes because of the simple construction processes and materials [9], [69]. Meanwhile, they are often located in complex scenes whose boundaries are difficult to recognize accurately [21]. Therefore, effective extraction of the shape and edge features is essential for improving the accuracy of illegal building identification. To realize this goal, we construct the Edge Flow to extract and integrate the shape and boundary information. Inspired by [70], we implement the GCL in the Edge Flow to catch the boundary information and enhance the ability of small object segmentation. The enhancement of small object segmentation is crucial for illegal building

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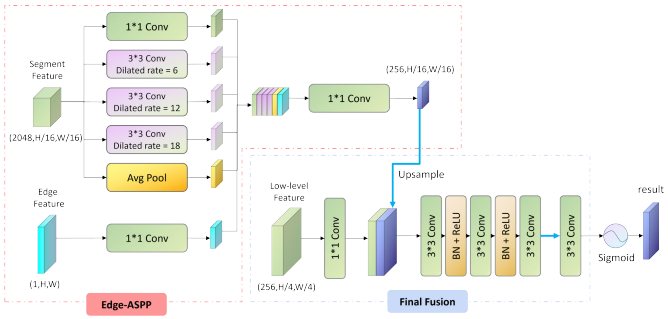


Fig. 6. Architecture of the Fusion Block. The blue arrows represent upsample layers.

detection since buildings are usually small as they are built by hand or with unsophisticated construction techniques. GCL is used in multiple locations in the Edge Flow, while the rest of this flow is ResNet Bottleneck blocks [64].

The structure of GCL is shown in Fig. 5. In GCL, we first generate an attention weight from the integration of the features of Segment Flow in the size of $C \times H \times W$ and the former gated edge features in the size of $1 \times H \times W$. The attention weight is computed as follows:

$$W_g = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_1\left(\mathrm{ReLU}\left(\mathrm{Conv}_1\left(\mathrm{BN}\left(F_i \| F_g\right)\right)\right)\right)\right)\right) \quad (4)$$

where σ represents the sigmoid function, F_i represents the input features generated from the Segment Flow, F_g represents the gated feature in the Edge Flow, and $\|$ represents channelwise concatenation. Then F_i and W_g are elementwise multiplied. The product of multiplication is processed with a residual connection and pointwise convolution to generate a new gated feature. The GCL structure incorporates complex semantic features from the Segment Flow into the Edge Flow, enabling the Edge Flow to extract complex edge information in a relatively simple network structure. Finally, the Edge Flow outputs an edge feature in the size of $1 \times H \times W$ supervised by the edge label of the ground truth.

3) Fusion Block: The Fusion Block consists of two parts which are Edge-ASPP and Final Fusion (Fig. 6). The Edge-ASPP is designed for fusing the deep semantic feature with the edge feature into multiscale features. The Final Fusion is used to further integrate the low-level features from the Segment Flow and the multiscale features generated by Edge-ASPP to output the final segmentation result.

The Edge-ASPP is adopted from the ASPP module [22]. First, the Edge-ASPP uses a 1×1 convolution layer to extract local information from the segment feature. Next, like ASPP, it uses 3×3 convolution layers with different dilated rates to extract information on different scales with different sizes of receptive fields. After that, an average pooling instead of a max pooling layer is utilized to get global information for robustness. Meanwhile, a pointwise convolution is used to reweight the edge features. Finally, all these features are concatenated and handled with a 1×1 convolution layer to build a multiscale feature with edge information.

The Final Fusion part uses several upsample and convolution layers to fuse the low-level feature from the Segment Flow and the multilevel feature from the Edge-ASPP to create the final change map. The low-level feature, which has only experienced two downsample operations in the Segment Flow, can bring more detailed information about high spatial resolution to the segmentation result.

B. Loss Function

We collectively train the three flows in the networks in an end-to-end fashion with two tasks. We supervise both the segmentation and the boundary results with the segment loss L_{seg} and the edge loss L_{edge} separately during the training. The total loss is the weighted sum of these two losses

$$L\left(y, \hat{y}, s, \hat{s}\right) = \lambda_1 L_{\mathrm{seg}}\left(y, \hat{y}\right) + \lambda_2 L_{\mathrm{edge}}\left(s, \hat{s}\right) \quad (5)$$

where y is the predicted label, $\hat{y}$ is the ground truth, s is the predicted boundaries, $\hat{s}$ is the edge label of the ground truth, and λ_1 and λ_2 are the weights of segment loss and edge loss separately.

In the context of illegal building detection and monitoring, the positive and negative samples in the dataset are highly imbalanced which considerably impacts the segmentation training [71]. To deal with the class imbalance, we employ a combination of two different loss functions including Focal loss [72] and Dice loss [73].

The Focal loss function reshapes the basic binary cross entropy function with the addition of a modulating factor $-\alpha(1-\hat{y})^{\lambda}$ which down-weights of easy negative and focuses on training on the hard negative examples. The Focal loss is defined as follows:

$$L_{\mathrm{focal}} = \begin{cases} -\alpha\left(1-\hat{y}\right)^{\lambda}\log\left(\hat{y}\right), & y = 1 \\ -(1-\alpha)\hat{y}^{\lambda}\log\left(1-\hat{y}\right), & y = 0 \end{cases} \quad (6)$$

where α and λ are tunable parameters that change the weight distribution in different sample scenarios.

The Dice loss function is a variation of the Dice coefficient, which could be defined as follows:

$$L_{\mathrm{dice}} = 1 - \frac{2|X \cap Y|}{|X| + |Y|} \quad (7)$$

where X and Y represent the ground truth and the predicted label, respectively. The Dice loss is a region-related loss function that considers loss information both locally and globally. Rather than the pointwise calculation of cross entropy loss, the Dice loss function considers the overlaps between the two sets: the ground truth and the predicted labels, thus being independent of the proportion of positive samples. Therefore, it offers an ideal solution to address the highly imbalanced data in image segmentation.

After obtaining the Focal loss and Dice loss, the segment loss is calculated in the form of a weighted sum of the two

$$L_{\mathrm{seg}} = \delta L_{\mathrm{focal}} + (1-\delta)L_{\mathrm{dice}} \quad (8)$$

where δ is the mixing weight of two losses. The binary cross entropy loss is used to supervise the edge labels, which is specifically described as follows:

$$L_{\mathrm{edge}} = -\left(y\log\left(\hat{y}\right) + (1-y)\log\left(1-\hat{y}\right)\right). \quad (9)$$

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Fig. 7. Research area of this work. Three sets of pictures on the left are samples of our datasets. The preconstruction image is on the left, and the postconstruction one is on the right. The ground truth is at the bottom of each image group. The projection of the map is UTM zone 50N, and the coordinate system is WGS84.

C. Dataset

The research area of this work is the areas inside the sixth ring road in Beijing, China. Beijing, as a representative Chinese city located in North China, exhibits high building densities in its urbanized region [74]. The region enclosed by the sixth ring road encompasses the majority of the urban and suburban areas surrounding the city center. The distribution of illegal buildings in this research area is characterized by sparsity (Fig. 7) and dispersion across urban and suburban regions. The presence of diverse urban forms [75], [76] and varying building density scenarios [74] within the research area enhances the generalizability of models trained in this context.

We use the VHR Google Earth Satellite image with a spatial resolution of 0.594 m and three bands (RGB) to construct the dataset for model training. DN values of these images have been uniformly calibrated to integers ranging from 0 to 255 by Google. The projection of the images is UTM zone 50N, and the coordinate system is WGS84. The selected satellite images cover the study area in 2019 and 2020. In light of the considerable uncertainty in the exact timelines of illegal building construction, we propose an inverse method to simulate the pre- and postconstruction scenarios of these structures. Specifically, we select a satellite image taken at the time of demolition as a representation of the preconstruction scenario, as both the preconstruction and postdemolition environments share the absence of the illegal building and similar surroundings. The resulting preconstruction (postdemolition) and postconstruction images are then paired and resized to 512×512 dimensions to form the bitemporal image dataset.

Labels of illegal buildings dismantled between 2019 and 2020 are provided by the Beijing municipal government. Additionally, we manually annotated labels of legal building changes in the test set to verify the model's ability to distinguish between illegal building and legal building changes. All building labels have been meticulously examined and revised to ensure precision and alignment with the Google Earth satellite images, specifically focusing on the accuracy of the edge labels

In total, the dataset consists of 5230 pairs of VHR images of preconstruction and postconstruction of illegal buildings. The bitemporal satellite images are divided into train, validation, and test sets with a ratio of 7:2:1. In total, 23 289 illegal building labels are included in the entire dataset, and 720 legal building labels are included in the test set.

Since the angles of images between pre- and postconstruction may vary, the coregistration of satellite images is implemented using AROSICS [77]. The image registration accuracy is guaranteed by AROSICS, whose registration accuracy increases as the spatial resolution of images increases [77]. Furthermore, histogram matching is applied for each bitemporal image pair to eliminate the effects of different lighting and weather conditions.

The sizes of illegal buildings in our dataset follow a heavy-tailed power law distribution [Fig. 8(d)] with most buildings having total areas less than 1000 m^2. For further understanding the shape characteristics of illegal buildings, several morphological metrics of the illegal buildings are computed. The rectangularity [78] describes the similarity of a polygon to a regular rectangle whose value equals the ratio of the area of a polygon to the area of its minimum bounding rotated rectangle. From our dataset, the rectangularity of most illegal buildings is higher than 0.8, indicating that illegal buildings tend to shape like rectangles [Fig. 8(a)]. The fractal dimension [79] evaluates the regularity of a polygon. A fractal dimension value close to 1 means that the polygon

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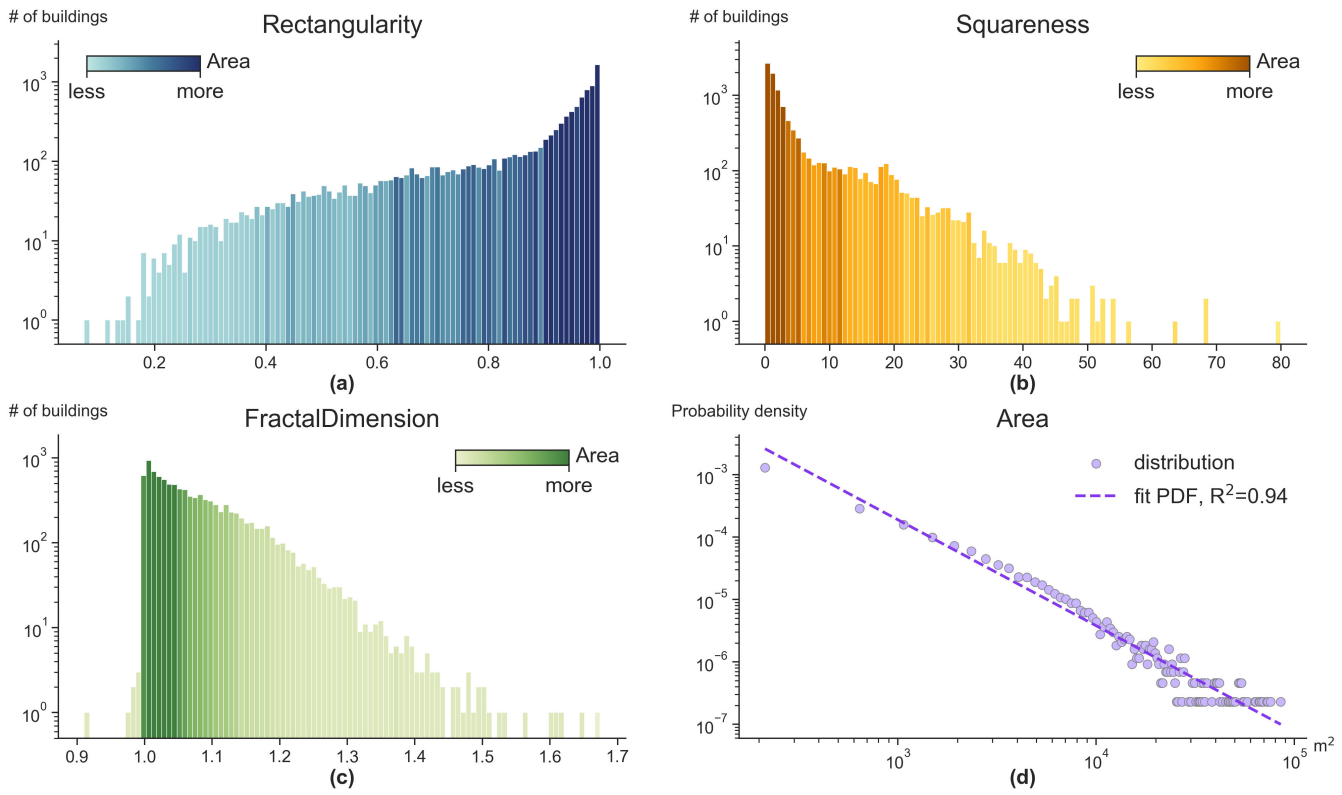


Fig. 8. Statistics of area and morphology metrics of illegal buildings. The darkness of colors in (a)–(c) represents the sum of buildings' area between bins. The dotted line in (d) is the fitted power law probability density function (pdf).

is completely regular. The illegal buildings in our dataset are shown to be highly regular, as most illegal buildings have fractal dimensions close to 1 [Fig. 8(b)]. Another morphological metric is the squareness [78] which evaluates the average deviation of angles of corners from 90° in a polygon. If a polygon has low squareness, its corners are close to orthogonal. The statistics in Fig. 8(c) indicate that most corners of illegal buildings are close to 90°. Overall, the illegal buildings in our dataset have small total areas, high rectangularity, are highly regular, and are close to being orthogonal.

IV. Experimental Results

A. Metrics

Several evaluation metrics of segmentation are used in this work for comprehensively evaluating the effectiveness of models, including precision (P), recall (R), overall accuracy (OA), and F1 score ($F1$). The specific descriptions of these metrics are shown in the formulas below

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{10}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{11}$$

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN}} \tag{12}$$

$$F1 = 2 \times \frac{P \times R}{P+R} \tag{13}$$

where TP represents the number of true positives, TN represents the number of true negatives, FP represents the number of false positives, and FN represents the number of false negatives.

In addition, to assess the ability of models in distinguishing between illegal and legal constructions without accessing information about legal building changes, we have developed several additional indicators for further evaluation. These indicators solely consider legal building changes as negative targets and illegal building changes as positive targets, while ignoring other pixels. This approach allows for the evaluation of the models' discriminative ability in identifying both illegal and legal building changes. We report the OA for Buildings (OA_B), the $F1$ score for Buildings ($F1_B$), and the Intersection over Union for Buildings (IoU_B) as comprehensive measures of the models' discriminative ability. The specific descriptions of these metrics are provided in the formulas below

$$\mathrm{OA}_B = \frac{\mathrm{TP}_B+\mathrm{TN}_B}{\mathrm{TP}_B+\mathrm{FP}_B+\mathrm{TN}_B+\mathrm{FN}_B} \tag{14}$$

$$F1_B = 2 \times \frac{P_B \times R_B}{P_B+R_B} \tag{15}$$

$$\mathrm{IoU}_B = \frac{\mathrm{TP}_B}{\mathrm{FP}_B+\mathrm{TP}_B+\mathrm{FN}_B} \tag{16}$$

where TP_B represents the number of true positives of illegal buildings in all building change pixels, TN_B represents the number of true negatives in all building change pixels, FP_B represents the number of false positives in all building change pixels, FN_B represents the number of false negatives in all building change pixels, $P_B = \mathrm{TP}_B/(\mathrm{TP}_B+\mathrm{FP}_B)$ represents the precision of illegal buildings, and $R_B = \mathrm{TP}_B/(\mathrm{TP}_B+\mathrm{FN}_B)$ represents the recall of illegal buildings.

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TABLE I
COMPARISON AMONG DIFFERENT NETWORKS ON IBMD

Metrics	P	R	OA	F1	OA_B	$F1_B$	IoU_B
FC-EF	0.7167	0.5597	0.9283	0.6285	0.5696	0.6859	0.5220
FC-Siam-conc	0.7525	0.7196	0.9460	0.7357	0.7038	0.8050	0.6736
FC-Siam-diff	0.7879	0.7096	0.9494	0.7467	0.7064	0.8060	0.6751
STANet	0.6026	**0.8168**	0.9239	0.6935	0.7596	**0.8563**	0.7335
SNUNet	0.7534	0.7090	0.9449	0.7306	0.7041	0.8065	0.6756
BIT	0.7779	0.7576	0.9553	0.7675	0.7373	0.8322	0.7126
ChangeFormer	0.7655	0.6940	0.9495	0.7280	0.6858	0.7909	0.6540
IBMNet	**0.8125**	0.7860	**0.9586**	**0.7990**	**0.7660**	0.8538	**0.7449**

B. Illegal Building Monitoring Results

1) Comparison Methods: To elaborate on the effectiveness of our illegal building identification method, we choose both classic as well as state-of-the-art change detection models to apply on the IBMD dataset for comparisons. For classic fully convolutional networks, we select FC-EF [37], FC-Siam-Diff [37], and FC-Siam-Conc [37]. On the other hand, STANet [80] and SNUNet [81] are chosen as advanced CNN-based change detection models. Moreover, BIT [80] and ChangeFormer [50] are selected as state-of-the-art transformer-based models. Briefs of these networks are listed below.

1) *Fully Convolutional Early Fusion (FC-EF) [37]:* A method that directly adds the bitemporal images followed by the change detection with a fully convolutional network.
2) *Fully Convolutional Siamese-Difference (FC-Siam-Diff) [37]:* A method that uses a weight-sharing Siamese encoder to process the bitemporal images and fuse the hierarchical feature with differentiation.
3) *Fully Convolutional Siamese-Concatenation (FC-Siam-Conc) [37]:* A method similar to the FC-Siam-Diff. Instead of subtracting, it concatenates feature maps generated in two encoder branches for multilevel feature fusion.
4) *Spatial–Temporal Attention Neural Network (STANet) [80]:* A method that uses contrastive learning to learn the difference between two images. The network uses multiple self-attention modules to extract multiscale features to generate better change maps.
5) *Siam-NestedUNet (SNUNet) [81]:* A method that employs NestedUNet with a double-branch encoder whose branches share the same weights. Its network structure enables deep mining of multiscale features. Additionally, a channel attention module is utilized in the decoder to achieve finer segmentation results.
6) *Bitemporal Image Transformer (BIT) [48]:* A method that introduces the transformer to the change detection task to handle with semantic information. This model first uses CNN-like ResNet to extract features from bitemporal images and then uses transformers to refine the image features and generate context-rich tokens. Finally, the model uses a decoder to generate the change map.
7) *ChangeFormer [50]:* A method similar to BIT, while exclusively relies on transformers to both extract features from bitemporal images and generate the final change maps.

2) Implementation Details: In this study, the divisions of training, validation, and test set are random and uniform in geographical distribution to minimize data bias. The data augmentation methods, including random gamma transform, random flipping transform, and random rotation, are performed in the training set to improve model generalization. The order of the training set is shuffled in every epoch.

Specifically, our model uses the Adam optimizer with a batch size of 16. The initial learning rate and weight decay are 10^{-4} and 10^{-3}, respectively. The step learning rate scheduler is used to decay the learning rate every 30 epochs by a factor of 0.1. Our proposed method has been trained on the IBMD for 80 epochs. In terms of loss functions, we manually set the loss weights λ_1 and λ_2 to 2 and 1, respectively. The mixing weight δ is set to 0.5. The training process is implemented on a server with 2 NVIDIA GeForce RTX Titan X GPU, and the model is implemented with PyTorch 1.12.0. During the training process, we use the $F1$ score to monitor the performance of the model. In the training set, this metric eventually oscillated around 0.87, while in the validation set, it oscillated around 0.8. The IBMNet as well as all other comparison models are trained by the same configurations for fair comparison.

3) Comparison Experiments on the Illegal Building Monitoring Task: Table I shows the metric scoring results of the comparison experiments among our IBMNet model and other benchmark models. The IBMNet achieves excellent results in various basic evaluation metrics, getting the highest score in P, OA, and $F1$ on the IBMD and reaching 0.7990 on the $F1$ score. Therefore, our proposed method could effectively detect and monitor the existing illegal buildings using bitemporal remote sensing images based on change detection. Compared to other evaluation metrics, the improvement in OA is relatively small for IBMNet, which can be explained by the fact that the OA is highly dominated by a large number of negative samples. Although the recall of IBMNet is in the second place and is slightly less than STANet, its substantial leads in other evaluation metrics confirm its relative advantages in illegal building monitoring.

In terms of the discrimination of illegal buildings and legal buildings, our model also demonstrates superior performance compared to other models. Specifically, the IBMNet achieves the highest scores in OA_B and IoU_B, reaching 0.7660 and 0.7449, respectively. Furthermore, the $F1_B$ score of IBMNet is 0.8538, which is the second highest among all the models, and only slightly lower than the highest score achieved by

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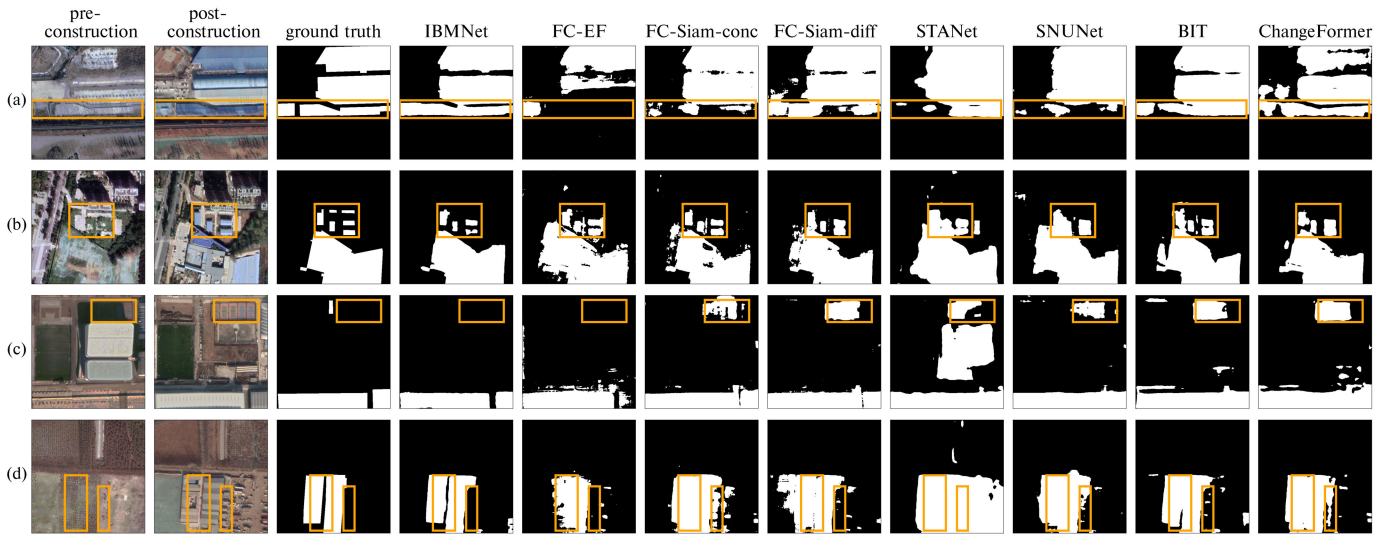


Fig. 9. Comparison of detection results among models. Information and details worth noticing are marked with orange boxes.

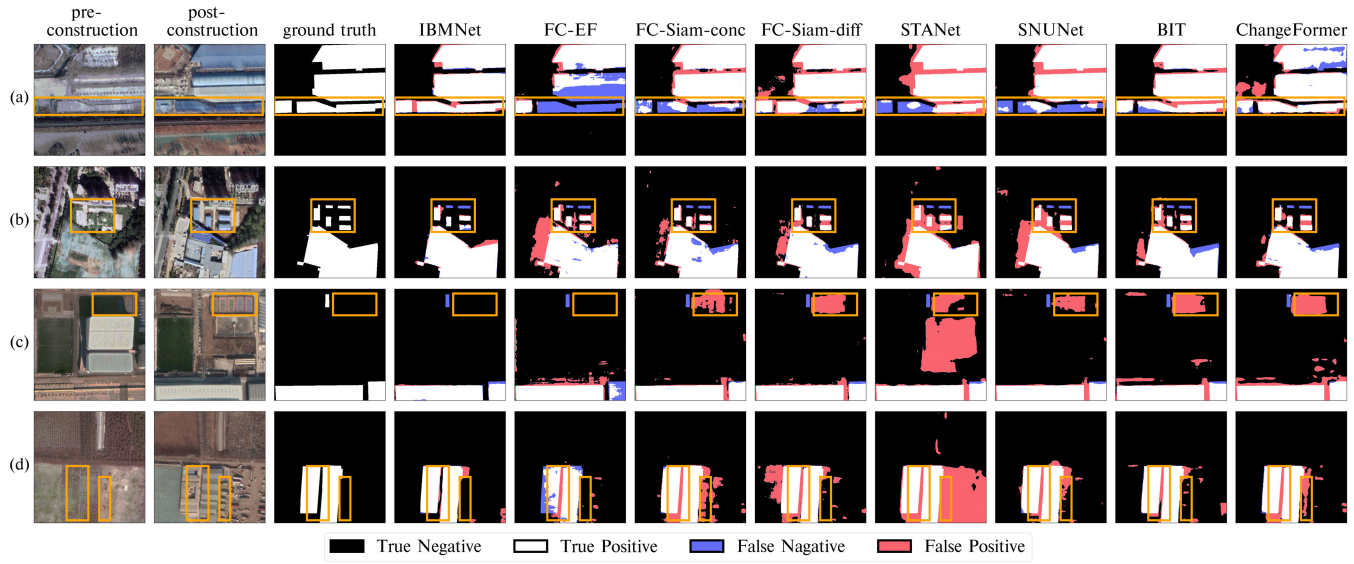


Fig. 10. Comparison of false labeling among models. False negative and False positive results are labeled by blue and red, respectively.

STANet. The high $F1_B$ of STANet could be attributed to the enhanced P_B of STANet caused by the omission of pixels labeled as no-change in calculation.

Fig. 9 shows the illegal building detection results of models in several different conditions. In general, the results of IBMNet have more clear edges and are less fragmented compared with other methods. Row(a) shows the models' ability to distinguish buildings from small color changes. The IBMNet outputs clear extractions of the building while other comparison models fail to generate outputs with compatible quality. Row(b) indicates the circumstance of detecting small buildings within the annotation box. We find that the IBMNet model generates the most precise boundary and distinguishable building targets among the six models. Row(c) manifests that our model can identify no-change pixels from shadows and distinguish the illegal buildings from other newly built legal buildings. Row(d) illustrates that IBMNet can avoid the impact of the object sharing similar features (like containers) around the illegal buildings on the segmentation boundaries. Additionally, both (a) and (d) show the ability of IBMNet to accurately split and extract features from densely distributed buildings.

Fig. 10 uses different colors to visualize the FN and FP, where FN is colored blue and FP is colored red. We can notice that the IBMNet has a solid capacity to depict the boundaries of illegal buildings. Compared with other models, the false label of our model surrounding the building edge is rarer. The overall amount of false positive predictions of IBMNet is also the smallest among all the models, suggesting splendid precision of model performance.

Fig. 11 illustrates the capacity of our model to differentiate between illegal and legal buildings. The two sets of images depict buildings that resemble illegal structures and are prone to misclassification. Set (a) demonstrates our model's proficiency in distinguishing illegal buildings from temporary constructions commonly found on construction sites. Of all

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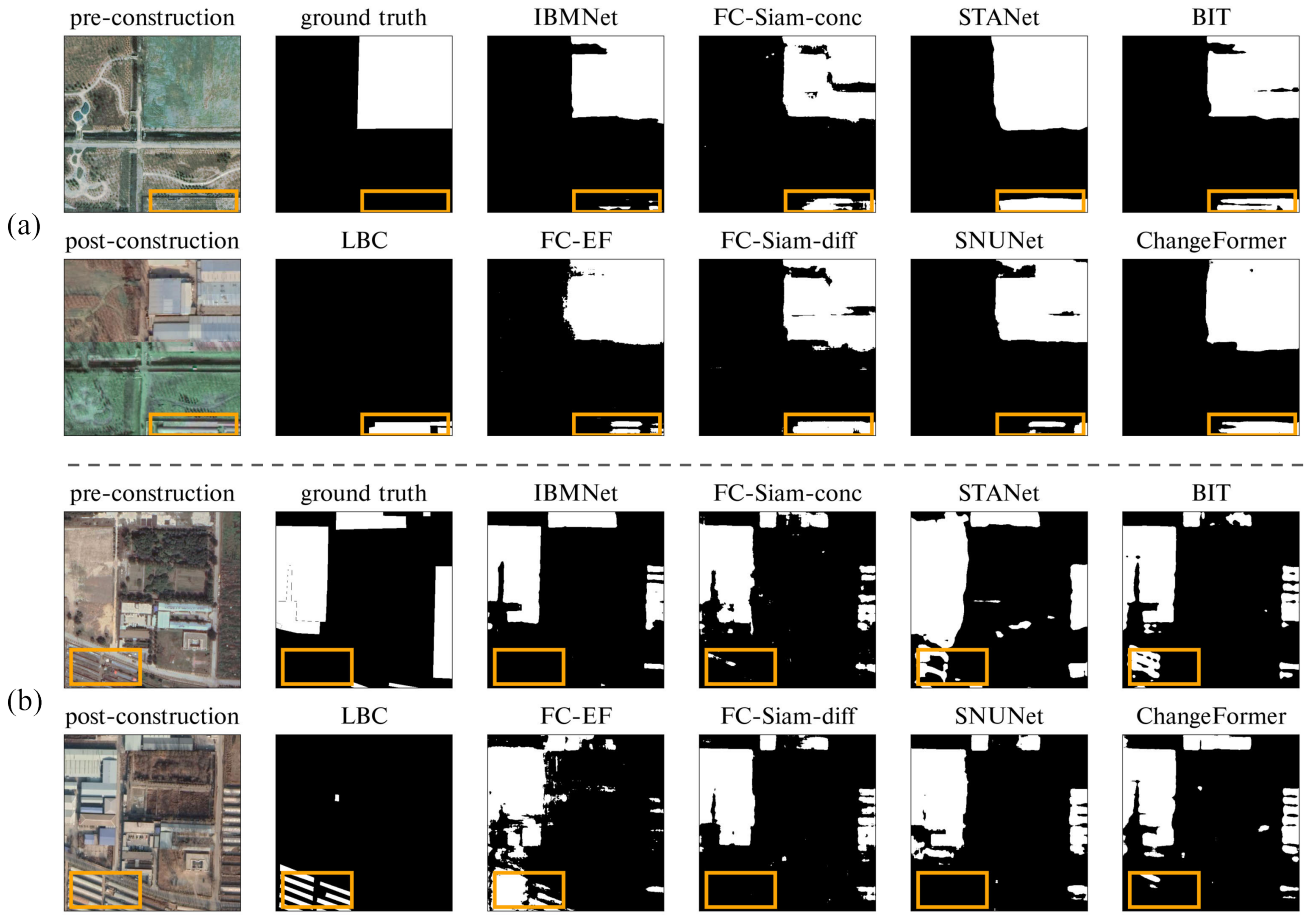


Fig. 11. Comparison cases of distinguish illegal buildings' change and legal buildings' change. LBC refers to the ground truth of legal building changes. The orange boxes mark the details worth noticing.

the models, IBMNet is the least impacted by the presence of legal temporary buildings in its detection results. Set (b) showcases our model's competence in distinguishing illegal buildings from seasonal structures such as greenhouses. While many other models misclassified, IBMNet correctly identifies greenhouses as legal buildings. Furthermore, our proposed method generates the most precise illegal building monitoring results among models that correctly classifies legal greenhouses. Overall, our method effectively discriminates illegal buildings from other types of structures and produces satisfactory results in illegal building detection.

V. Discussion

A. Ablation Studies

In this article, we apply ablation studies on our model by changing or removing the parts of the network to evaluate the effects of different modules. The basic network includes the Siamese encoder and ASPP while the inclusion of Edge Flow, AFM, and different fusion operators of AFM are tested. In Table II, the comparison among experiments (1)–(3) indicates the impact of the proposed Edge Flow and edge loss on the result of illegal building monitoring. Experiments (3) and (5) demonstrate that the AFM is effective in improving model performances. Additionally, the results of experiments (4)–(6) verify that our combination method is effective and reasonable.

The comparison between experiment (3) and experiment (5) demonstrates the effectiveness of AFM. With the same fusion operators, experiment (5), added with the AFM, has approximately 2% increases in recall and precision compared to experiment (3). The comprehensive metric $F1$ and discriminative metric IoU_B also show increases in experiment (5).

The comparison among experiments (4)–(6) indicates that neither concatenation nor differentiation alone is the best way to fuse the features in AFM at the first step. Properly designing fusion methods and reasonably combining the two fusion operators can contribute to the optimal solution. For the features subsequently processed by the Edge Flow, it is plausible to use differentiation for better detection of boundaries of the building changes. On the other hand, the concatenation is favored for the features fed into the Fusion Block to retain more information. Overall, the ablation experiments verify the reasonability of our model design which consists of Edge Flow, AFM, and hybrid fusion operations in terms of several evaluation metrics.

B. Effectiveness of the Edge Flow

We use gradient-weighted class activation mapping (GradCAM) [82] to visualize the attentional heatmaps for the three GCLs in the Edge Flow for further analyzing its effectiveness. GradCAM uses the gradients in the networks to calculate the

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TABLE II
QUANTITATIVE EVALUATION RESULTS OF ABLATION STUDIES

No.	Edge Flow	AFM	Fusion Op.	Loss	P	R	OA	F1	OA_B	$F1_B$	IoU_B
(1)	×	×	Concat.	Segment loss	0.6882	0.6999	0.9342	0.6940	0.6786	0.7882	0.6505
(2)	√	×	Concat.	Segment loss	0.7152	0.7537	0.9402	0.7339	0.7220	0.8242	0.7009
(3)	√	×	Concat.	Segment loss & Edge loss	0.7744	0.7734	0.9527	0.7738	0.7488	0.8424	0.7278
(4)	√	√	Diff.	Segment loss & Edge loss	0.7541	**0.8005**	0.9520	0.7766	0.7627	0.8514	0.7413
(5)	√	√	Concat.	Segemnt loss & Edge loss	0.7917	0.7999	0.9571	0.7957	0.7603	0.8512	0.7410
(6)	√	√	Concat. & Diff.	Segment loss & Edge loss	**0.8125**	0.7860	**0.9586**	**0.7990**	**0.7660**	**0.8538**	**0.7449**

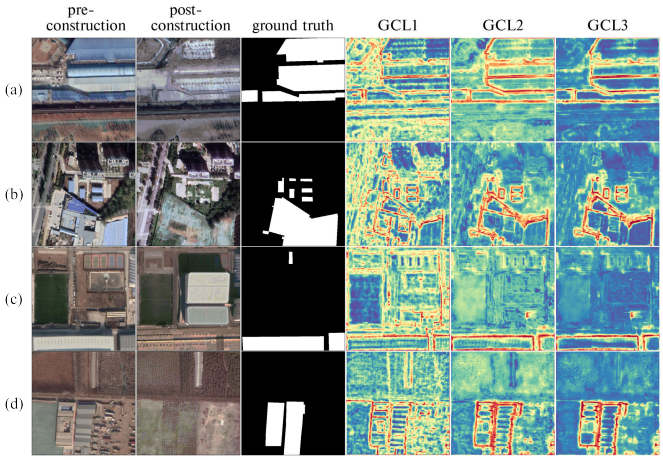


Fig. 12. GradCAM of 3 Gated Convolutional Layers. GCL1, 2, and 3 denote the three gated convolutional layers of the Edge Flow in turn. Areas focused by convolutional layers are shown in red while the unfocused areas are shown in blue.



Fig. 13. Comparison between IBMNet with and without AFM.

class activation map (CAM) of a specific layer, reflecting the layer focus point in inferencing. Fig. 12 presents the samples of image pairs, ground truths, and the GradCAM of three GCLs. From Fig. 12, it is worth noticing that the three GCLs have different focuses. The first GCL mainly focuses on shallow and explicit edges in both images such as the border of trees, roads, and roofs. As the network goes deeper, the Edge Flow pays more attention to these significant edges that help refine the boundary of correct segmentation results. The second GCL reduces the focus on unimportant shallow boundaries and focuses more on the deep edges of the scene such as the boundaries between different buildings, buildings and roads, buildings and soil, and buildings and similar containers. The third GCL further adjusts the attention and widens the attention gap between important edges and other parts of the scene. Comparisons of the same samples in Figs. 9 and 12 show that the Edge Flow is a valid approach for confining the boundaries of buildings. The GCL's ability in focusing on the important edges makes the IBMNet to generate more accurate segmentation results compared to other models, as displayed in Fig. 9.

C. Effectiveness of AFM

We also apply the GradCAM to evaluate the effectiveness of AFM. In Fig. 13, we compare the fusion results of models with and without AFM. For a more robust comparison, we choose the results with the same fusion operator which is the concatenation method for both networks. From Fig. 13, we can find that the AFM makes the network more focused on valuable changes and less susceptible to the interference of environments and other pseudo-changes. Compared with models with AFM, the model without AFM is more affected by the pseudo-changes caused by color differences which are caused by factors such as light conditions and shadowing.

VI. CONCLUSION

In this study, we propose IBMNet, a novel approach for the timely monitoring of illegal buildings based on change detection of VHR satellite imagery to improve the capability of information collection for illegal construction. We designed a two-flow network to focus on edge information and improve the accuracy of results. Meanwhile, the IBMD was built for the training and evaluation of IBMNet. Additionally, we also propose several metrics to assess the model's ability in distinguishing legal and illegal buildings, including $F1_B$, OA_B, and IoU_B. Our model achieved the highest performances in both illegal building detection metrics ($F1 = 0.7990$) and building discrimination metrics ($IoU_B = 0.7449$) on the IBMD dataset among all the selected models in the study. The visualization results indicate that IBMNet can detect illegal buildings on various backgrounds and distinguish between the construction of illegal buildings and legal buildings. Our proposed method can be implemented at a higher frequency than the methods based on urban databases, enabling the timely discovery of illegal buildings and saving government costs in handling and demolishing. Additionally, IBMNet can be used on a larger scale compared to the existing methods based on urban datasets and can cover suburban areas where the coverage of urban datasets is incomplete.

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However, there are still limitations to our research. Obtaining actual data on illegal buildings is difficult, so we were only able to obtain data on illegal buildings in a single city for one year to train and validate our model, limiting the validation of its transferability. Some illegal buildings may be too small to be distinguished even in VHR images, making it difficult for both humans and neural networks to extract useful features from only a few pixels. Furthermore, the surveillance of illegal construction is complex in practice, and our work can only assist governments in terms of data collection of illegal buildings rather than fully solve the problem. Some challenges may arise during the practical implementation of city governance, primarily stemming from government-imposed restrictions on device limitations, network security, and data confidentiality.

In summary, we conducted a preliminary verification of the effectiveness of our model in Beijing, China. Considering the shared characteristics of buildings, the potential for its application in other cities and countries is also promising. In areas where illegal structures resemble those found in Beijing, primarily consisting of simple or colored steel buildings, the model may be readily transferred and implemented without the need for additional training. However, in situations where the building styles and characteristics significantly differ from those in Beijing, fine-tuning may be necessary to adapt to local conditions. Additionally, our model also has the potential to be further applied in other targets that are similar with illegal buildings in characteristics or scales, such as photovoltaic panels, greenhouses, and color steel houses.

In future work, we will continue to conduct more experiments in other cities and other countries for further evaluation of the feasibility and effectiveness of our model. Additionally, we are planning to further working on the classification of illegal buildings along with the monitoring to provide more detailed information for city governance.

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